

DSA Practical No: 02

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Div: SY3

Class: C3

INPUT :

```
#include <stdio.h>
```

```
int binarySearch(int arr[], int low, int high, int key)
```

```
{
```

```
    int mid;
```

```
    if (low > high)
```

```
        return -1;
```

```
    mid = (low + high) / 2;
```

```
    if (arr[mid] == key)
```

```
        return mid;
```

```
    if (key < arr[mid])
```

```
        return binarySearch(arr, low, mid - 1, key);
```

```
    return binarySearch(arr, mid + 1, high, key);
```

```
}
```

```
int main()
```

```
{
```

```
    int arr[100];
```

```
    int n, i, key, result;
```

```
char ch;
```

```
do
```

```
{
```

```
    printf(" RECURSIVE BINARY SEARCH\n");
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter %d sorted elements:\n", n);
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    printf("Enter element to search: ");
```

```
    scanf("%d", &key);
```

```
    result = binarySearch(arr, 0, n - 1, key);
```

```
    if (result == -1)
```

```
    {
```

```
        printf("Element %d is not found in the array.\n", key);
```

```
    }
```

```
    else
```

```
    {
```

```

        printf("Element %d is found at index %d.\n", key, result);
    }

    printf("\nDo you want to continue? (y/n): ");
    scanf(" %c", &ch);

} while (ch == 'y' || ch == 'Y');

printf("\nProgram ended.\n");

return 0;
}

```

OUTPUT :

```

PS D:\dsa\Experiment-02-Recursive-Binary-Search> .\program.exe
    RECURSIVE BINARY SEARCH
Enter number of elements: 5
Enter 5 sorted elements:
1
2
3
4
5
Enter element to search: 2
Element 2 is found at index 1.

Do you want to continue? (y/n): y
    RECURSIVE BINARY SEARCH
Enter number of elements: 5
Enter 5 sorted elements:
5
4
3
2
1
Enter element to search: 2
Element 2 is not found in the array.

Do you want to continue? (y/n): 

```